

Code

Automatic Dustbin's Code:

```
#include <Servo.h>
```

```
Servo servo;
```

```
int trigPin = 9;
```

```
int echoPin = 8;
```

```
long duration;
```

```
int distance;
```

```
bool lidOpen = false;
```

```
unsigned long lastSeenTime = 0;
```

```
unsigned long waitTime = 3000;
```

```
void setup() {
```

```
    servo.attach(7);
```

```
    servo.write(0);
```

```
    pinMode(trigPin, OUTPUT);
```

```
    pinMode(echoPin, INPUT);
```

```
Serial.begin(9600);  
}  
  
void loop() {  
  digitalWrite(trigPin, LOW);  
  delayMicroseconds(2);  
  digitalWrite(trigPin, HIGH);  
  delayMicroseconds(10);  
  digitalWrite(trigPin, LOW);  
  
  duration = pulseIn(echoPin, HIGH);  
  distance = duration * 0.034 / 2;  
  
  if (distance <= 6) {  
    servo.write(90);  
    lidOpen = true;  
    lastSeenTime = millis();  
  }  
  
  if (lidOpen && distance > 6) {  
    if (millis() - lastSeenTime >= waitTime) {  
      servo.write(0);
```

```
    lidOpen = false;
}
}

delay(100);
}
```

Human Following Car's Code:

```
const byte ENA = 9;
const byte IN1 = 5;
const byte IN2 = 6;

const byte IN3 = 7;
const byte IN4 = 8;
const byte ENB = 10;

const byte IR_LEFT = 11;
const byte IR_RIGHT = 2;

const byte TRIG_PIN = 4;
const byte ECHO_PIN = 3;
```

```
int speed = 150;
int minDist = 10;
int maxDist = 40;

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);
  pinMode(ENB, OUTPUT);

  pinMode(IR_LEFT, INPUT);
  pinMode(IR_RIGHT, INPUT);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  Serial.begin(9600);
}
```

```
void loop() {  
    int distance = getDistance();  
  
    int leftIR = digitalRead(IR_LEFT);  
    int rightIR = digitalRead(IR_RIGHT);  
  
    Serial.print("Distance: ");  
    Serial.println(distance);  
  
    if (distance > maxDist) {  
        stopCar();  
    }  
    else if (distance > minDist && distance <= maxDist) {  
  
        forward();  
    }  
    else {  
        stopCar();  
    }  
}
```

```
if (leftIR == LOW && rightIR == HIGH) {  
    turnLeft();  
}  
else if (leftIR == HIGH && rightIR == LOW) {  
    turnRight();  
}  
  
delay(100);  
}
```

```
int getDistance() {  
    digitalWrite(TRIG_PIN, LOW);  
    delayMicroseconds(2);  
  
    digitalWrite(TRIG_PIN, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(TRIG_PIN, LOW);  
  
    long duration = pulseIn(ECHO_PIN, HIGH);
```

```
int distance = duration * 0.034 / 2;
```

```
return distance;
```

```
}
```

```
void forward() {
```

```
    analogWrite(ENA, speed);
```

```
    analogWrite(ENB, speed);
```

```
    digitalWrite(IN1, HIGH);
```

```
    digitalWrite(IN2, LOW);
```

```
    digitalWrite(IN3, HIGH);
```

```
    digitalWrite(IN4, LOW);
```

```
}
```

```
void stopCar() {
```

```
    digitalWrite(IN1, LOW);
```

```
    digitalWrite(IN2, LOW);
```

```
    digitalWrite(IN3, LOW);
```

```
    digitalWrite(IN4, LOW);
```

```
}
```

```
void turnLeft() {  
    analogWrite(ENA, speed / 2);  
    analogWrite(ENB, speed);  
  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, HIGH);  
  
    digitalWrite(IN3, HIGH);  
    digitalWrite(IN4, LOW);  
}
```

```
void turnRight() {  
    analogWrite(ENA, speed);  
    analogWrite(ENB, speed / 2);  
  
    digitalWrite(IN1, HIGH);  
    digitalWrite(IN2, LOW);  
  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, HIGH);  
}
```